

Automotive

ROHM offers comprehensive support with a complete array of products optimized for the automotive market

Air conditioner | Instrument/ AC panel | Rain sensor | Mirror ECU | Body control | TPMS | Air bag | ABS | Engine control | EPS | Camera Module / Network | Stop lamp

Pb Free — Halogen Free & Antimony Free

RoHS — Directive Compliance

IC Product Specifications



Voltage Trackers

Part No.	Offset Voltage (mV)	Output Current (A)	Maximum Voltage(V)	Circuit Current (μA)	Operating Temperature(°C)
New BD3925FP	±15 (Ta=-40 to +125°C, Vcc=6.36V, Io=0.200mA)	0.5	50	45	-40 to +125
New BD3925HFP	±15 (Ta=-40 to +125°C, Vcc=6.36V, Io=0.200mA)	0.5	50	45	-40 to +125

High-reliability Serial EEPROMs

Part No.	Density (bit)	Supply Voltage Range (V)	Operating Temperature (Range)(°C)	Lifetime (Erase) (Times)(25°C)	Data Retention (yrs) (Times)(85°C)
BR25HCC□0 series	1K to 32K	2.5 to 5.5	-40 to +125	10^6	40
BR93HC□ series	2K to 16K	2.7 to 5.5	-40 to +125	10^6	40

High Voltage LDO Regulators

Part No.	Output Voltage (V)	Output Voltage Precision (V)	Output Current (A)	Maximum Voltage(V)	Circuit Current (μA)	Operating Temperature(°C)
New BD3570FP	3.3	±3 (Ta=-40 to +125°C)	0.5	50	30	-40 to +125
New BD3570HFP	3.3	±3 (Ta=-40 to +125°C)	0.5	50	30	-40 to +125
New BD3571FP	5.0	±3 (Ta=-40 to +125°C)	0.5	50	30	-40 to +125
New BD3571HFP	5.0	±3 (Ta=-40 to +125°C)	0.5	50	30	-40 to +125
New BD3572FP	Variable	±3 (Ta=-40 to +125°C)	0.5	50	30	-40 to +125
New BD3572HFP	Variable	±3 (Ta=-40 to +125°C)	0.5	50	30	-40 to +125
New BD3573FP	3.3	±3 (Ta=-40 to +125°C)	0.5	50	30	-40 to +125
New BD3573HFP	3.3	±3 (Ta=-40 to +125°C)	0.5	50	30	-40 to +125
New BD3574FP	5.0	±3 (Ta=-40 to +125°C)	0.5	50	30	-40 to +125
New BD3574HFP	5.0	±3 (Ta=-40 to +125°C)	0.5	50	30	-40 to +125
New BD3575FP	Variable	±3 (Ta=-40 to +125°C)	0.5	50	30	-40 to +125
New BD3575HFP	Variable	±3 (Ta=-40 to +125°C)	0.5	50	30	-40 to +125

Regulators with Voltage Detector and Watchdog Timer

Part No.	VDD				Reset			
	Vcc (Max.)(V)	Vcc (Min.)(V)	Io (Max.)(A)	Io (Limit)(A)	Vdet (Min.)(V)	Vdet (Typ.)(V)	Vdet (Max.)(V)	Vdet (Hys.)(%)
BD3004HFP	5	±2 (Ta=-40 to +125°C)	0.5	0.78 (Io=-200mA)	Variable (at Vi open : 4.1V)		±2	Adjustable Voltage (external resistor)
BD3006HFP	5	±2 (Ta=-40 to +125°C)	0.5	0.78 (Io=-200mA)	4.5		±2	SCI Voltage Detect +VDT(Vdet-vdet)
New BD3010AFV	5	±3 (Ta=-40 to +125°C)	0.2	0.55 (Io=-150mA)	Variable (at Vdet open : 4.25V)		±3	Adjustable Voltage (external resistor)

Flexible Step-down Switching Regulators with Built-in Power MOSFET

Part No.	Supply Voltage (V)	Output Current (A)	Output Voltage(V)	Reference Voltage Accuracy (%)	Operating Temperature(°C)	Oscillation Circuit
BD9001F	7 to 48	2.0	1 to Vin	±2	-40 to +85	Self-oscillation
BD9778F	7 to 35	2.0	1 to Vin	±2	-40 to +125	Self-oscillation
BD9778HFP	7 to 35	2.0	1 to Vin	±2	-40 to +125	Self-oscillation
BD9781HFP	7 to 35	4.0	1 to Vin	±2	-40 to +125	Self-oscillation (Selectionchronizable)

Automotive Grade Operational Amplifiers/Comparators

Operational Amplifiers: Ground Sense

Part No.	Number of Circuits	Supply Voltage (V)	Circuit Current (mA)	Input Offset Voltage (mV)	Slew Rate (V/μs)	Gain Bandwidth (MHz)
New BA2904HFVM-C	2	3 to 32	0.7	2	0.2	0.5
New BA2902HFV-C	4	3 to 32	0.7	2	0.2	0.5

Comparators: Ground Sense

Part No.	Number of Circuits	Supply Voltage (V)	Circuit Current (mA)	Input Offset Voltage (mV)	Input Bias Current (nA)	Output Current (mA)	Input Voltage Range
New BA2903HFVM-C	2	2 to 36	0.6	2	50	16	VCC to VCC+1.5
New BA2901HFV-C	4	2 to 36	0.8	2	50	16	Vee to Vcc-1.5

LED Driver IC for Indicators

Part No.	Power Supply (V)	Output Current (mA)	Number of Inputs	Number of Outputs	Output Type	Maximum Voltage(V)
BD8105FV	4.5 to 5.5	50	5	12	Open Drain	35

Discrete Product Specifications



MOSFETs

Part No.	Package	Polarity	Absolute Maximum Ratings (max.)				Qg (nC)	Driving Voltage(V)		
			Vdss (V)	Id (A)	Pd (W) (Tc=25°C)	Vgss (V)	Vds (V)	Vgs (V)		
RSS090N03	SOP8	Nch	30	9	2	17	15	11	11	4
RSS085N05	SOP8	Nch	45	8.5	2	18	16	13	15.3	4
RSS065N06	SOP8	Nch	60	6.5	2	31	28	24	11	4
RSS075P03	SOP8	Pch	-30	-7.5	2	25	22	15	30	4
RSS060P05	SOP8	Pch	-45	-6	2	38	35	26	23	4
SP8K3	SOP8	Nch+Nch	30	7	2	20	25	17	8.4	4
SP8J1	SOP8	Pch+Pch	-30	-5	2	43	40	30	16	4
SP8M3	SOP8	Nch	30	5	2	58	52	36	5.9	4
		Pch	-30	-4.5		65	57	40	8.5	4
RSQ045N03	TSMT6	Nch	30	4.5	1.25	40	36	27	6.8	4
RSQ035P03	TSMT6	Pch	-30	-3.5	1.25	70	65	45	9.2	4

Transistors for Solenoid Motors

Part No.	Package	Polarity	Vcc (V)	Ic (A)	Pc(mW) (Ta=25°C)	Qd	R1(kΩ) (typ)
DTDG14GP	MPT3	NPN	60±10	1	500	300 or more	none/10
DTDG23YP	MPT3	NPN	60±10	1	500	300 or more	2.2/10
2SD2143	CPT3	NPN	60±10	2	1000	1k to 10k	3.5/0.3
QSH29	TSMT6	NPN+NPN	60±10	0.5	500	500 or more	none/10

Rectifier Diodes

Part No.	Package	Vrm (V)	Io (A)	Ifsm (A)
RR254M-400	PMDU	400	0.7	35
RR274EA-400	TSMD5	400	0.5	8

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